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Will the Market Deliver? A Business Ethics Perspective to AI

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00:03

Hello, my name is Christoph Lütge. I am a professor of business ethics at the Technical University of Munich. And I'm also the director of the TUM Institute for Ethics in Artificial Intelligence. Today, I want to talk about business ethics perspective on AI and of course, ethical issues in AI in particular. Let me share my screen. Business ethics perspective is something that, as I believe, is quite important to the entire field of AI and ethics. And it is my impression that so far, in many cases, this business ethics perspective has been lacking. AI ethical issues have centered around issues that have something to do with technical issues with data protection and others, but often the perspective of a corporation, of companies, large companies, small companies, and also, the question of customers and markets is something that is not often found in this discussion. And that's why we believe we need to change that. So, what I will talk about today is, I will give some introduction to AI and to ethics in general, I will talk about how we generate fairness and trust in AI systems. I will talk about how we can define responsibility and liability in the AI systems, something that is very important for corporations. I will talk about the risks that AI poses, in particular to companies. I will talk about psychological aspects, privacy, evaluating the AI technology, and also a bit about military applications.

02:14

So let me start with some general remarks on AI about definitions, concepts, and applications. But what is artificial intelligence, and there are a number of definitions around here. One of them goes back to Alan Turing, who is seen here, who said that a machine can be called intelligent when it is able to exhibit a behavior that cannot be distinguished from the behavior of an intelligent human being. And several other definitions go in this direction and tying the concept of artificial intelligence relatively close to the human intelligence. But this can be debated. Is artificial intelligence something that is or should be measured by the standards of human intelligence? Or is it something entirely different? I believe the latter, I believe that it's better to think of AI in some different way. Since an AI may exhibit certain features of human intelligence, which are easily seen, but on the other hand, exhibit features that might be quite different. And you might underestimate artificial intelligence, if you just try to measure it with human standards. That's why I like this last definition that is used here by Haenline and Kaplan, according to which AI is the system's ability to interpret external data correctly, to learn from such data and to use those learnings to achieve specific goals and tasks through flexible adaptation. So really

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these three components together make for a meaningful definition of AI, a one that is not necessarily tied to what human intelligence can do. That's why we also should not just think of it in this way. For example, if you use these famous cases of AlphaGo. AlphaGo was the machine that beat a professional human player in Go was very surprising to those human beings to those professional players. They didn't think that it was possible that a machine could learn through such flexible adaptation and in a very fast period of time. When you compare this to the DeepBlue case many years ago, almost 20 years before AlphaGo that was an entirely different type of AI. And it can also be questioned whether that is the kind of AI something that I will not go into. But certainly, there is quite a difference between the types of chess computers of the 90s say, which used rather a brute force type of AI than the current systems which are based on deep learning, on machine learning, the AlphaGo systems and others. So, according to our book "An Introduction to Ethics in Robotics and AI", which you can also find in open access on the internet. According to this are the most basic element common to all the definitions of AI is that artificial intelligence involves the study design, and building of intelligent agents that can achieve goals. Just briefly, we of course, distinguish between narrow AI and general AI. So narrow AI, those are systems that are related to specific questions and which involve machine learning tailored to specific tasks, such as face or speech recognition, or automated translations. That is something we definitely have, we do not yet have this second type, the general AI with which we usually designate systems that would decide proactively and include following elements, logical reasoning and deliberation, decisions under uncertainty, and the ability to learn on a scale that would combine many different skills in a way that an artificial AI system would be almost equivalent to a human being on many levels and that we are far from it.

07:19

Some of the fields of AI use about which I will be talking in this lecture are telemedicine. So the use of tech AI technologies in the field of healthcare, the health and healthcare medicine, the care robots, of course, in care homes, autonomous driving, drugs, and vaccine development, an important study, especially now, and the area of sustainability. But before doing that, let me just give some remarks on AI ethics and its benefits and challenges in general. What is the public perception on AI? I have collected some headlines here for you, there you can see that there are already both challenges as well as ethical opportunities present in these. Of course, we find headlines like "We Teach AI systems Everything, Including our Biases". So here, it's already mentioned that buyers, potential buyers of these systems is an ethical problem, or the "Dangerous Rise of Military AI: Machines That Lead to Slaughter", this is a very dramatic headline. But on the other hand, we also see that "Cutting Edge Robotic Surgery Helps Combat COVID" in this case, or "AI is Revolutionizing" and I take it that this is meant in a positive way here, human resource functions. All this leads to the conclusion that AI is a global challenge as we know, but also AI ethics is one, and that is something that companies are increasingly recognizing.

09:30

Because from an ethical point of view, it's clear that many instruments that we have developed can be used for good or for bad purposes. We can use a knife to cut the cake into pieces or we can use it to kill people. And that is the same with AI. So the use of AI depends on the circumstances of its use, of the user, and the consequences of course for uninvolved parties. When it comes to business ethics, what is the kind of framework in which we can operate here? In general, the use of technologies such as AI

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involves risks which need to be considered and weighed against each other and weighed also against the opportunities that come along with it. So, we need a kind of evaluations team or a cost benefit analysis, not just material costs, but also other costs. And one framework to assess the effects of AI are the UN Sustainable Development Goals, which are pictured here. This is a framework that has gained so much recognition and attention during the last years that it's quite, quite standard to suppose it. There are a number of ways of approaching AI in this context. For example, AI creates an added value for, and with value I mean ethical value, for the following Sustainable Development Goals, such as smart grids for examples, the use of smart grids or intelligent technologies for current and for reducing CO2 emissions. For example, this is something that is in the Sustainable Development Goals 11 and 13. We also can use AI with regard to SDGs 9 and 17 when it comes to global integration, reduction in spatial distances. When it comes to women's rights, there are chat bots that are seen as instruments for empowering women. This is with reference to SDG 5. Or just to take some examples here, the court decisions, this is about is SDG 10 and 16. Investment decisions of banks and financial institutions SDG 8 or a very important one, the transformation of agriculture and the use of AI and big data technologies in agriculture. This even involves three SDGs, 2, 16 and 15. So, there are a lot of general benefits, but there are also some more specific ones. For example, AI, as I just mentioned, is an important element or technology that can help in efficient food production. It can transform production by monitoring, managing, in a better way environmental conditions and crop yields. It can help reduce fertilizer use and water waste. There are some companies that can be named in this sector. This is a project in the same similar area that is being pursued by my Institute here. In when it comes to AI for Earth observation AI for EO, there it is quite helpful. It is quite helpful to use AI for Earth observation technologies. It can help for example, in creating planning tools that capture the relevant feedback dynamics of the water supply systems prevailing in, for example, slum areas, it can help form the basis of an infrastructure plan. And it will search in order to expand certain capacities of infrastructure into the STEM areas in a relatively inexpensive way for eventually improving our water supplies.

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The second point is about telemedicine, telemedicine is a technology which can be quite helpful for combating certain problems of many of our societies such as the rural population, and the rural services that are not always readily available. There are uneven distributions of physicians for example, telemedicine can be a solution to reach patients in those remote areas. And it might also at least during the times of a pandemic to help avoid unnecessary contacts. The care robots - care robots can help those societies which have to deal with the problems of aging societies and address the shortage that we face in terms of qualified healthcare personnel. So we have robots specifically designed for a number of these care tasks. And some are saying well, these might even be more humane to use, at least for certain tasks. And especially older people find them more helpful in these areas so that they can for other tasks rather use human beings which would then have more time. So human healthcare personnel. So it's not just a question of machine versus man, it's just where it's more of a question where do you use the machine so that human beings can also be more both efficiently as well as ethically be assigned. Surgical robots will certainly see a huge rise in the years to come, they improve the surgeons abilities to create precise and minimally invasive incisions, stitch wounds and so on. And here as well as many other areas, they will mainly be used to assist surgeons, but of course, this would gradually shift to a more and more even if in the near future, probably not fully autonomous use. AI can

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also help in diagnosing a COVID there are already AI solutions for reading COVID scan or CT scan images that exhibit certain features of COVID-19 infections or diseases. And it can also help for developing drugs that can be used for COVID, but also for other diseases. So, for COVID we have already developed a number of vaccines around the globe fortunately, but for other diseases in the future, AI and big data solutions can be very helpful for crunching the huge numbers, that need to be dealt with in order to speed up the development of vaccines process. For example, the MIT and IBM collaborate on an AI to assist physicians to determine how long in terms of care patients need to receive ventilator support. So, even beyond the development of vaccines, it can be used in the treatment of sick people.

17:56

An entirely different field where also ethical AI can be helpful is the field of autonomous driving. And of course, this is very, very interesting for car companies, it will be a huge market to some extent with assisting systems. It is already and they are of course, for companies it's very important to see how can we bring people to trust those systems and that will only be possible through the development of ethical AI. So giving those systems certain limits, certain rules, certain programming, which will lead to people trusting them. And that will be an ethical opportunity in itself because it will help to bring the high fatality rates down that we currently have in all countries around the world. And we see the same statistics everywhere that at least 90% of car accidents are caused by human error. And if autonomous cars can help save many lives here and reduce crashes and reduce injuries, then this will be an ethical enterprise in itself.

19:32

One very controversial use of AI usually is the use of AI in the courts. Something that is as I said it's controversial the number of countries are not using it yet and others do. Here as well as other areas we must always ask - is the AI better in comparison to the human alternative? Because of course, we cannot just completely take the AI and say, well, we don't want these results, we don't want those results. We have to compare it to the human performance. And just to cite one example from the courts is that the study that looked at death sentences handed out in the US, over the course of more than 30 years, researchers found that more than 4% of these candidates were indeed innocent. And so, we can infer from that human beings make mistakes. Or if you look at, for example, detecting cancer, for a long time, it has had been said, well, AI systems are good at detecting cancer, but on the other hand, human beings, experienced human beings, experienced doctors, can do better, but that has changed. So whatever the case about the death sentence may be, but in terms of cancer recognition, there now, recent studies that were published in 2019 and 2020, determined that AI is better than any human being. And the question then arises, so is it ethically mandatory to use AI? Given that it performs better than human beings? Should we implement AI under the condition that it is better than human beings? Or do we need to consider additional aspects?

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So far, I have talked about a number of ethical opportunities, but of course, there are ethical challenges as well. First, there's the question of the dependence on the accuracy of technical systems. Those examples that I mentioned from telemedicine, from autonomous driving, and also in other areas, make

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it clear that these systems need to be very, very precise before we can accept them and trust in them. Then there is the question of security of these systems, how vulnerable are they against attacks from the outside, against hacking attempts? And another ethical challenge is the question of privacy and the danger of data misuse. So in Europe, we have developed the general data protection regulation to assess this problem of privacy. However, it will not solve all problems that arise in connection with AI. We face other problems such as biases, biases of AI, which I mentioned in the beginning, fairness of AI and non-discrimination, that is something that is not straightforward, as well as the general problem of digital literacy.

23:07

So for example, when we come to a challenge that is probably one of the most pressing ones for companies, so in the context of business ethics, this is one that I'm also regularly asked to give my opinion about. What is the responsibility and what is the liability of companies in cases where AI has done wrong, such as, as we see here, when autonomous vehicles crash? And we have had the first examples of such crashes. We have seen the famous, to the left we have an example from Google a number of years ago, we have the Uber self-driving car that hit a pedestrian that was jaywalking on the street. And the report eventually found that there's no basis for criminal liability against Uber, as well as the Tesla crash from years ago, where the the autopilot failed to detect a truck and Tesla then said well, the customer is always in control and responsible for what happened. But that will not suffice for the future. So, with the current changes that are underway, it will be companies that will and must be held liable for problems that arise with the use of AI. And especially in the case of autonomous driving, if a car has been in autonomous mode during a crash then the individual driver cannot be held responsible, and the current changes in regulation that are underway also go into this direction. So we will see companies being held liable, also in the sense of strict liability, so that you don't have to prove in each case what happened, but in general for the use of their system. So the type of product liability as we also see in many other areas. One example here is the use of killer robots. And of course, the use of killer robots is, if you use that term, is a very wide topic, I will just limit myself to the point how to distinguish between what is in this context, considered a legitimate target, as well as an illegitimate target. For example, what if the child that with a weapon or maybe just with a toy weapon is considered an enemy? And we will see those those cases if we use these systems. What would that mean for a company that manufactures those weapons? And certainly, if this weapon was in autonomous mode, then the company must be held liable for such damages and collateral damages.

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I mentioned the case of biases and fairness. What if a company or if an AI system that a company uses is found to be discriminating against a certain part of the population, for example, in terms of race. If we find that, for example, a widely used algorithm, as in this case, shows a significant distortion in its results, that lead to an underestimation of the health risks of African Americans? And that is something that can damage both company's reputation, as well as leading to illegal fines. Here in this case, the bias arose because of the training data, the training data that the system was used with predicted, I mean, they were based on the past, and so they could only learn from what they had. So they were trained to predict the health costs, and not the diseases., but unequal access to this healthcare system meant that as a rule, in this case, black patients spent less money on care than white patients. What

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this means is that, well, quality matters and companies have a self-interest in trustworthy AI. And in this specific case, it's important to look at your training data. What training data do you use? What biases might be in this training data? And do you need additional data maybe, such as data that might be also constructed, maybe specifically for this purpose? And that is a huge task for the future and many companies are already working on this problem. We see this in other areas which we'll just briefly mention, the famous Amazon case where recruiting was biased against women, as well as prison sentences where AI is being helped to determine the length of a prison system, but also, we need to make sure that we do not just extend our human biases into the AI world. So for summing this up, we see both opportunities as well as challenges in this area. And this is very relevant for companies. So while certainly AI can help us to enable our self-realization, to enhance human agency, and increase our societal capabilities, it can also remove our responsibilities, reduce our human controls or devalue our skills, that is something companies need to consider. How can they consider it?

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So, one important step is to look at AI governance and first principles for AI governance, and principles for these and then further rules have been developed in the past years. I will start with a non-European or non-Western example - the Beijing AI principles, which were developed in 2019. And there's a huge overlap between these and others that I will get to in a minute. So, there are principles that say do good, be responsible, control risks, use AI wisely, informed consent as an important aspect, also education of people, and governance principles like adaptation, long-term planning, and so on. In Europe, the European Commission has put out an approach to AI governance that focuses on an ecosystem of excellence, which I will skip, but if you are interested in that, just look more into these details. It has laid out the first ever legal framework for AI and in 2021, this is one of the first important steps that we have done in general on the planet. So, this legal framework distinguishes between two types of risks, one with regard to AI, one that is called unacceptable risks - a clear threat to the safety, livelihoods and rights of people, such AI systems should it says should be banned, as well as a high-risk, high-risk that where AI is used in critical infrastructures, transport, employment, law enforcement, and such AI systems will be subject to strict obligations before being put on the market. So there we will need adequate risk assessments and mitigation systems also with regard to the quality of the datasets. There is a coordinated plan with member states of the EU to create leadership in trustworthy AI. This also by the way, means that there are also a number of systems which do not pose a risk and which can then be widely used without much action.

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This is based on some thoughts from the Data Ethics Commission in Germany. However, there is a certain distinction between this and the European Framework, I will just skip this for the moment and rather come to the ethical principles behind it. And the ethical principles were developed by especially these two think tanks - the AI4People which I was a member, as well as the High Level Expert Group on AI through 2018 mainly and they were set up in this system of trustworthy A. It requires accountability, I would just mentioned this one, we have diversity also, we have this the question of non-discrimination but fairness though, there's again the question of biases, there is the societal environmental well-being so, the aspect of sustainability is there. But as I mentioned, accountability is a key aspect that many companies are interested in and the new regulatory framework that has started

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now and will be developed further is an important element for all markets. And we have certain technical as well as non-technical methods for ethical AI. These have been recommended by the High Level Expert Group in the EU. And I will just mention translating the trustworthiness into procedures. This is very important. So trustworthy AI is the general principle, the general idea but to translate these into procedures is of course the task for the future. Testing and validating the models keeping, them robust and stable as much as possible, documenting decisions. There's the question of auditability and also explaining those decisions to end users but also to people in the process. That is an entire field of its own explainable AI research. We also have non-technical methods such as well, regulation, as I already mentioned, but also standardization, developing accountability governance frameworks for organizations, we have developing codes of conduct for example, those types of codes that are already being used in fields like Corporate Social Responsibility can be extended and used. We need to see more stakeholder dialogue with members and organizations of civil society. And the companies should also involve and look at the question of diversity of their teams. So if you want to avoid discrimination, have a look at your teams, your development teams, let's see what you can do there. And all this is summed up into five general ethical principles, beneficence, so do good, non-maleficence - do no harm, autonomy - strike a balance between decision making by machines as well as human beings, justice - the benefits of those systems should be shared by all, and finally, explicability - a particular ethical principle for AI systems, which are often seen as a black box and companies will need to look into how to create interfaces. As I already mentioned, this is a technical and non-technical task to create these interfaces, so that people can have at least a basic understanding of what and why the results came about of a certain AI system.

36:42

To end with an example from the automotive sector, I had the pleasure of chairing the AI4People Automotive Committee throughout 2020, where we developed ethical guidelines for the automotive sector, together with industry experts and researchers from fields of ethics, law, engineering, technology and others. So, there we discussed issues, which were both ethical as well as technological ones, technological robustness and safety, but also privacy, data governance, non-discrimination, and many others. For example, in this case, we came to a consensus on three points in our Committee, that it should be permitted at any time for autonomous vehicles to balance the risks. So, not just to put one risk above any others, but to balance all those risks. Also, we argue for a step-by-step approach for the introduction of autonomous vehicles. So, not just take all the steps at once and come up with level four which is nearly full autonomous driving in one step, but do this in an incremental way. And also develop clear regulatory framework, so that companies can have a safe background for which they can develop and also put to use their technologies that they have. These are the details of that and I encourage you to have a look at our article.

38:46

So to conclude from a business ethics point of view, but also from a general ethical perspective, AI is a dual use, it can be used for good and bad purposes. For good purposes include achieving something in terms of UN Sustainable Development Goals. I mentioned the example of care robots, also fighting diseases, but bad purposes might include no clear cut litigation, I mean, this is not something that we should aim for, but something that we should avoid, there must be a clear liability. If we use military AI,

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then we must make sure also to give them certain ethical rules and clear regulatory framework, as well as biases in algorithms That is something we need to avoid and that is a risk that many companies face and might also be suitable. So to ensure ethical instead of destructive AI, especially for companies and corporations, we need exchange between industry and research, as well as civil society, I would add in politics of course. We need to update our regulatory frameworks such as in the recently published rules of the European Commission. We need practical guidelines for ethical AI for example, in the forms of tools for companies, that is something that a number of institutions such as the IEAI around the world is currently doing. And we need to develop trust in AI and thereby we need to focus on meeting societal acceptance, because ultimately, as I like to say, AI cannot fly without ethics. AI without ethics is not possible because without ethics, without the trust in these systems, people, users, consumers will just not use these solutions, these systems. Thank you very much.